

Jihwan Moon

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Summary

- Applied Scientist at INFOCZ Inc. and Ph.D. candidate (**coursework completed**) at Seoul National University's Vision and Learning Lab, advised by Prof. Gunhee Kim; work spans **data-centric AI research** and **large-scale production ML**, from problem formulation and modeling to deployed systems.
- Build the ML core of **GOS**, an enterprise cybersecurity product that processes **1.4B+ provenance-graph edges per day** of EDR telemetry and is being deployed to **10K+ hosts at Samsung Electronics**, scaling toward **200K**; owned its inference and serving path for a **10×+ speedup**.
- Published data-centric AI research at top venues: online video–text data filtering (**CVPR 2025**, first author), sample selection under noisy labels (**NeurIPS 2024**), and decoupled greedy block-wise learning (**ICLR 2021**).
- My research aims to **make AI more accessible and affordable**: local (greedy block-wise) learning and decentralized learning for LLM pretraining and fine-tuning, and meta-learning for generalizable ontology-construction harnesses.

Professional & Research Experience

INFOCZ Inc.

Seoul, South Korea

Applied Scientist

Aug 2026 – Present

- Lead research on graph neural network architectures and self-supervised learning for advanced persistent threat (APT) detection in **GOS**, an enterprise cybersecurity product for large-scale EDR telemetry that combines provenance-graph modeling, GNN-based anomaly detection, and LLM-assisted filtering and investigation.
- Contributed to the provenance-graph processing and ML pipeline behind GOS — security-log ingestion, graph construction, feature generation, and anomaly detection at **1.4B+ edges per day** — now being deployed to **10K+ hosts at Samsung Electronics**, with a planned rollout to **200K hosts**.
- Helped raise Specificity @ Recall = 1.0 from **69.5% to 99.7%** (GNN only) and **99.98%** (GNN + LLM filter); the system surfaced **18 attack behaviors** missed by the incumbent security system, spanning Lateral Movement, Discovery, Reconnaissance, Persistence, Command & Control, and Defense Evasion.
- First-authored an arXiv preprint auditing evaluation in provenance-based intrusion detection, the problem setting behind GOS: audited **nine** released implementations, ran **four** detectors with their original code, isolated three measurement effects behind reported scores, and proposed **STRICT**, six checkable reporting requirements.
- Optimized model inference and serving for GOS by designing a producer–consumer pipeline for better CPU–GPU overlap and scheduling, achieving a speedup of over **10×**.
- Lead research on meta-learning for generalizable ontology-construction harnesses.

KC ML2

Seoul, South Korea

Research Resident

Jun 2026 – Aug 2026

- Led research on local (greedy block-wise) learning for LLM pretraining, reducing the memory and interconnect requirements of large-model training.
- Submitted a paper to the NeurIPS 2026 CODEC-FM workshop.
- Co-authored a hierarchical windowed roto-reflection equivariant ViT for equivariant feature extraction, accepted to the NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (Extended Abstract Track).

LG AI Research

Seoul, South Korea

Research Scientist Intern, Bio Intelligence Lab

Sep 2025 – Mar 2026

- Researched token compression and pruning for Vision Transformers processing gigapixel-scale inputs, including whole-slide images (WSIs), to cut the compute cost of full-resolution inference.

Allen Institute for AI (AI2)

Research Collaborator

Jun 2024 – Feb 2025

- Co-led online video–text data filtering research with AI2 researchers for efficient training on web-scale data streams; published at CVPR 2025.

Vision and Learning Lab, Seoul National University

Seoul, South Korea

Graduate Research Assistant (Ph.D. coursework completed)

Mar 2021 – Present

- Conducted and led research on multimodal data filtering, noisy-label sample selection, and improving multimodal LLMs with a multimodal knowledge base (NeurIPS 2024, CVPR 2025).

Undergraduate Research Intern

Jul 2020 – Feb 2021

- Searched for optimal neural network block configurations for greedy block-wise (local) learning, matching or beating backpropagation with asynchronous block training; published at ICLR 2021.

Platform Inc.

Seoul, South Korea

NLP / Deep Learning Engineer

Jun 2019 – Jan 2020

- Improved a production sticker and emoji recommendation system using NLP and deep learning.
- Won **1st place** in the EmotionGIF 2020 challenge.

Publications

Where the Numbers Come From: Auditing Evaluation in Provenance-Based Intrusion Detection

arXiv 2026

Jihwan Moon, Gunhee Kim, Myeongjang Pyeon

Refined Annotations and Specialized Augmentation for Minor Epicardial Arteries in Coronary Angiography

Scientific Reports 2026

Do-Hyun Kim, Sangwoo Moon, Daehyeon Choe, Je Yoon Shin, Su Jin Kim, Jae Jin Lee, Daehong Kang, *Jihwan Moon*, Si-Hyuck Kang

Hierarchical Windowed Roto-reflection Equivariant ViT for Equivariant Feature Extraction

NeurIPS 2026 Workshop

Sheir A. Zaheer, *Jihwan Moon*, Chan Y. Park

Workshop on Symmetry and Geometry in Neural Representations (Extended Abstract Track)

ReSpec: Relevance and Specificity Grounded Online Filtering for Learning on Video-Text Data Streams

CVPR 2025

Jihwan Moon^{*}, Chris Dongjoo Kim^{*}, Sangwoo Moon, Heeseung Yun, Sihaeng Lee, Aniruddha Kembhavi, Soonyoung Lee, Gunhee Kim, Sangho Lee, Christopher Clark

Sample Selection via Contrastive Fragmentation for Noisy Label Regression

NeurIPS 2024

Chris Dongjoo Kim^{*}, Sangwoo Moon^{*}, *Jihwan Moon*, Dongyeon Woo, Gunhee Kim

SEDONA: Search for Decoupled Neural Networks Toward Greedy Block-wise Learning

ICLR 2021

Myeongjang Pyeon, *Jihwan Moon*, Taeyoung Hahn, Gunhee Kim

^{*} *Equal contribution.*

Education

Seoul National University

Seoul, South Korea

Integrated M.S. & Ph.D. program in Artificial Intelligence

Mar 2021 – Present

Ph.D. coursework completed

- Advisor: Prof. Gunhee Kim; GPA: 4.12/4.3

Seoul National University

Seoul, South Korea

B.S. in Economics

Mar 2013 – Feb 2021

- Minor in Computer Science; GPA: 3.67/4.3

Skills & Languages

Research: Data-Centric AI (data selection & filtering, noisy labels), Graph Neural Networks, Anomaly Detection, Self-Supervised Learning, Multimodal Representation Learning, Meta-Learning, Efficient Training (local / block-wise, decentralized), LLM Pretraining & Fine-tuning

Programming: Python, Java **Frameworks:** PyTorch, TensorFlow

Systems: Inference & serving optimization (CPU–GPU overlap, scheduling), large-scale graph data pipelines, distributed training

Languages: Korean (native), English (fluent)

Academic Service & Teaching

Reviewer

- Journal: IEEE TPAMI (2024, 2025)
- Conferences: CVPR, ECCV, NeurIPS (2026)

Teaching Assistant (Seoul National University)

- Probabilistic Graphical Model (Spring 2022, Spring 2025)
- Discrete Mathematics (Spring 2023)
- Theory and Lab of IoT, AI and Big Data (Fall 2022)